University of Otago
ŌTĀKOU WHAKAIHU WAKA

SCHOOL OF COMPUTING

COSC343

Artificial Intelligence

Semester 2

Lecturers

Dr. Lech Szymanski
Office: Owheo 2.32
Email: lech.szymanski@otago.ac.nz

Dr. Veronica Liesaputra
Office: Owheo 2.21
Email: veronica.liesaputra@otago.ac.nz

Lectures

2×1h per week

Labs

1×2h per week (Streamed)

Course Description

In this paper we will look at different definitions of intelligence and at the concept of intelligent agents, concentrating on the issue of how to get information about the world and how to make use of it. We will consider techniques for machine learning and probabilistic reasoning. Almost every human ability results from learning from experience: we will look at how these learning processes can be modelled computationally.

Topics covered include:

- search and optimisation algorithms (including genetic algorithms);
- probabilistic reasoning methods (including Bayesian methods);
- machine learning algorithms (with a focus on neural networks).

Learning Outcomes

By completion of this paper students are expected to:

- Understand fundamental concepts relating to classical Artificial Intelligence methods.
- Develop appreciation for challenges of developing robust and reliable AI acting in the real world
- Have fundamental understanding of a range of AI methods and their applicability for different situations.
- Implement a basic algorithm from each of the different approaches listed above on a toy example, including but not limited to: simple games, basic spam filters, identifying objects in an image.
- Become adept with usage of basic software libraries for general machine learning (i.e. Python + sklearn)
- Develop awareness of ethical issues related to increasing pervasiveness of AI in everyday life along with the impact and repercussions of use of self-learning/adaptable systems.

Lecture/Lab Schedule

Week	Lecture	Lab
1	Introduction	Robot navigation
	Agents and environments	
2	Uninformed search	8-puzzle
	Informed search	
3	Adversarial search	TicTacToe
	Stochastic search	
4	Uncertainty	Catch up/8 queens
	Bayesian reasoning I	
5	Bayesian reasoning II	Spam filtering
	Learning	
6	Classification I	Iris data
	Classification II	
7	Regression I	Regression
	Regression II	
8	Multilayer perceptron I	Catch up/8 queens
	Multilayer perceptron II	
9	Multilayer perceptron III	Digits
	Simple recurrent neural network	
10	Reinforcement learning I	Frozen lake
	Reinforcement learning II	
11	Unsupervised learning I	Data analysis
	Unsupervised leanring II	
12	Ethics I	Text generation
	Ethics II	
13	Review	

Labs

Laboratory exercises provide opportunities for developing practical understanding and implementation of methods that complement the theory covered in the lectures. All the programming will be done in Python. The development environment consist of VSCode running Anaconda virtual environment Python interpreter.

Check your eVision timetable to find the time and location of your lab stream.

Course Readings

Prescribed Course Textbook:

Stuart J. Russel, Peter Norvig, *Artificial intelligence : a modern approach* (2020), 4th edition, Pearson, Boston.

The book provides extended reading for almost every lecture, as well as comprehensive coverage of related AI topics. It is strongly recommended that you purchase or borrow this book for the semester.

Course Administration

The paper will be administered through Aoroa.

Course Workload and Expectations

This is an 18-point paper. For your guidance, we offer the following breakdown of hours:

Activity	**Hours**	**Details**
Lectures	26 hours	(2h per week for 13 weeks)
Labs	26 hours	(2h per week for 13 weeks)
Studies/Reading	48 hours	(4h per week for 12 weeks)
Assignments	80 hours	(for two assignments over 9 weeks)
Total	180 hours	

Students are expected to attend lectures. While handouts summarising the lecture slides will be provided, keeping good notes is recommended. The lectures will be recorded and available on Echo360 afterward, but these recordings will capture limited aspects of the lecture, such as the slides and the voice-over. The main purpose of these recordings is to serve as a supplement and a review of the lectures, and they should be viewed as a replacement only if necessary for those who cannot attend sporadically (due to sickness, etc.).

Students should prepare for labs by studying the class notes and reviewing the lab script. A demonstrator will be present in the lab to assist everyone in completing the exercise.

Assessment

Internal assessment	**Due**	**Worth**
Labs		**10%**
Assignment 1	Week 5 of the course	**20%**
Assignment 2	Week 9 of the course	**20%**
Exam		50%
Total assessment		**100%**

Mandatory Pass Requirement

To pass this course, you must earn at least 20 of the 50 course marks available from the final examination, regardless of your performance in other assessments.

Assignment 1 (20%)

Students will write a program in Python to implement an agent for a simple task, such as playing a simple game. The framework for the environment will be provided. Students will evaluate their agent and assess its effectiveness. The objective of this assessment is not necessarily to implement any specific method taught in the course, but rather to encourage students to develop their own strategies for creating an intelligent software solution for a given problem.

Students will be assessed on:

- the effectiveness of their agent in solving a given problem;
- the quality of the written code;
- the clarity of the technical report explaining methodology and the results.

Assignment 2 (20%)

Students will write a program in Python to implement a game that involves a population of agents. This time, the agents will need to learn successful strategies through a genetic algorithm. The environment will be provided, and students will design their agent, fitness function, and all aspects of the genetic algorithm that governs its learning.

Students will be assessed on:

- the effectiveness of their agent in solving a given problem;
- the quality of the written code;
- the clarity of the technical report explaining methodology and the results.

Submitting Assignments

All assignments must be submitted electronically via Aoroa.

Assignment Returns

It is University policy that assignments must be marked, graded, and returned to students within three weeks of the due date.

Students will be informed in lectures when assignments will be ready and when they will be available for viewing on Aoroa. Please note that late assignments and those with approved extensions will be returned three weeks from the date that they were submitted.

Extensions, Late Work and Deductions

To obtain an extension, you must contact the course coordinator before the assignment due date. Except in cases of illness supported by a medical certificate, you will be required to provide evidence that you have already made progress on the assignment. Any work submitted after the final due date, or after the last date of an approved extension, will be considered late

Late work will incur a penalty of 5% (of the total mark) per day, including weekends. Assignments submitted more than five days after the due date, without an approved extension, will not be accepted – except in cases of illness or exceptional circumstances beyond the student's control. In such cases, you must contact the course coordinator as soon as possible and provide appropriate supporting evidence.

Academic Integrity and Academic Misconduct

Academic integrity means being honest in your studying and assessments. It is the basis for ethical decision-making and behaviour in an academic context. Academic integrity is informed by the values of honesty, trust, responsibility, fairness, respect and courage. Students are expected to be aware of, and act in accordance with, the University's Academic Integrity Policy.

Academic Misconduct, such as plagiarism or cheating, is a breach of Academic Integrity and is taken very seriously by the University. Types of misconduct include plagiarism, copying, unauthorised collaboration, taking unauthorised material into a test or exam, impersonation, and assisting someone else's misconduct. A more extensive list of the types of academic misconduct and associated processes and penalties is available in the University's Student Academic Misconduct Procedures.

Use of generative software such as ChatGPT is also considered academic misconduct unless specifically specified otherwise in the assignment handout. If generative software is used, students must specify how it was used and on what aspects of the assignment.

It is your responsibility to be aware of and use acceptable academic practices when completing your assessments. To access the information in the Academic Integrity Policy and learn more, please visit the University's Academic Integrity website or ask at the Student Learning Centre or Library. If you have any questions, ask your lecturer.

- Academic Integrity Policy
- Student Academic Misconduct Procedures

Support Services

Class representatives. Volunteers for class representative will be registered at the beginning of this course. The class representative is intended to help facilitate staff-student communication, enabling liaison, consultation and passage of information between teaching staff and the student body. Representatives usually meet with representatives of the academic staff twice a semester for about one hour. A full job description for the class representative is available from the Otago University Student's Association.

University Library. The Library website, `https://www.otago.ac.nz/library`, gives online access to Library services and resources including databases, past exam papers, referencing guides and the booking system for group study rooms.

Support for international students: `https://www.otago.ac.nz/international/support-services`.

Disabilities information and support: `https://www.otago.ac.nz/disabilities`.